

TASK 3

Write a C/JAVA program to implement encryption technique using Blowfish algorithm

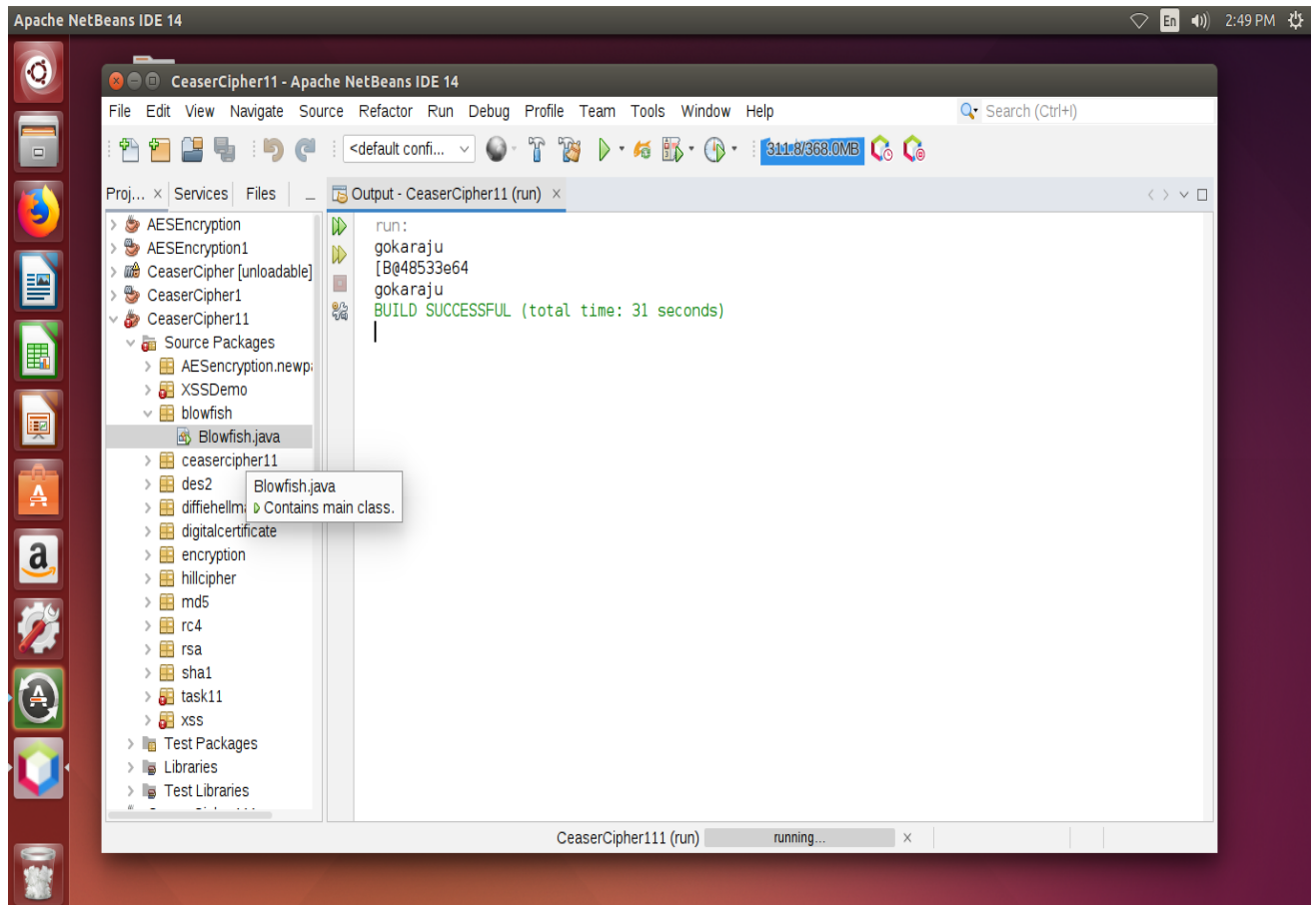
Aim: To write a Program to implement Blowfish algorithm logic.

Program:

```
import java.security.*;
import java.io.*;
import javax.crypto.*;
import java.util.*;

//DES
public class Task3 {
    static Scanner s = new
    Scanner(System.in); private static Cipher
    cipher;
    private static byte[] encrypt(String data) throws Exception{
        byte[] encrypted= cipher.doFinal(data.getBytes());
        System.out.println(encrypted);
        return encrypted;
    }
    private static byte[] decrypt(byte[] data) throws Exception{
        byte[] decrypted= cipher.doFinal(data);
        System.out.println(new
        String(decrypted)); return decrypted;
    }
    public static void main(String[] args)
    {
        try
        {
            String data = s.nextLine();
            KeyGenerator keygen = KeyGenerator.getInstance("Blowfish");
            SecretKey key = keygen.generateKey();
            cipher=Cipher.getInstance("Blowfish"
            );
            cipher.init(Cipher.ENCRYPT_MODE
            ,key); byte[] encrypted =
            encrypt(data);
            cipher.init(Cipher.DECRYPT_MODE,key);
            decrypt(encrypted);
        }
        catch (Exception e)
        {
            e.printStackTrace();
        }
    }
}
```

OUTPUT:



VIVA QUESTION

1. How does the Blowfish encryption process work, and what makes it secure?
2. What are the advantages and disadvantages of using Blowfish compared to AES and DES?
3. How does the Blowfish key expansion process contribute to its security?
4. Why is Blowfish not recommended for modern applications despite its strong security features?